UNIT 1: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

PREVIOUS YEAR QUESTION (PYQ) BANK & MARKS-SCALED MODEL ANSWERS

SVKM's NMIMS MPSTME | Mumbai University | GTU | MDU Pattern (2012–2026)

EXECUTIVE OVERVIEW & SYLLABUS BOUNDARY AUDIT:
This manual compiles every Unit 1 term-end examination question extracted page by page from uploaded question papers (SVKM's NMIMS MPSTME, Mumbai University, GTU, and MDU). All questions preserve exact original wordings, marks, paper codes, months, and years. Questions testing identical concepts under different wordings are grouped into unified model answers scaled strictly to mark requirements (2–3 Marks, 4–5 Marks, 7–10 Marks) with indicative marking schemes and vector diagrams. Unit 2 topics (BFS, DFS, A*, Hill Climbing, Minimax) are strictly excluded per *Artificial Intelligence 2026-27.pdf*.

Year-Wise Exam Paper Distribution & Audit Summary

Exam Session / Paper	University / Body	Questions Extracted	Key Unit 1 Topics Tested
Nov 2022 Final Exam	NMIMS MPSTME	2 Questions (20 Marks)	5 Agent Types, Wumpus World PEAS
Jan 2023 Re-Exam	NMIMS MPSTME	2 Questions (14 Marks)	Environment Types, ML vs DL
Nov 2023 Final Exam	NMIMS MPSTME	2 Questions (15 Marks)	Medical PEAS, Learning Agent Architecture
Nov 2024 Final Exam	NMIMS MPSTME	2 Questions (10 Marks)	Learning Agent, Self-Driving Car & Doctor PEAS
Feb 2025 Re-Exam	NMIMS MPSTME	2 Questions (10 Marks)	Goal-Based Agent, Smart Refrigerator PEAS
Nov 2025 Final Exam	NMIMS MPSTME	2 Questions (15 Marks)	Learning Agent Architecture, Drone PEAS
Jan 2026 Re-Exam	NMIMS MPSTME	2 Questions (15 Marks)	Goal-Based Agent, Smart Refrigerator PEAS
May 2013 - Dec 2016	Mumbai University	12 Questions (55 Marks)	Turing Test, Rationality, Agent Types, PEAS
Winter 2012 - Summer 2020	GTU / MDU	9 Questions (38 Marks)	AI Definitions, Characteristics, Turing Test, PEAS

1.0 TOPIC 1: DEFINITIONS, HISTORY, FOUNDATIONS & 4 APPROACHES TO AI

PYQ GROUP 1.1: Definitions of AI and Characteristics of AI Problems

Repetition Frequency: *Asked 5 times (GTU 2016, 2019, 2020 | MDU 2015, 2018)*

- *"Define AI? Explain the characteristics of AI problem."* — **GTU Winter-16, Q.2** [7 Marks]
- *"What is Artificial Intelligence? Explain the applications areas of Artificial Intelligence in detail."* — **MDU BCA 2018, Q2.b** [5 Marks]
- *"What is AI? Explain its origin and its various techniques."* — **MDU BCA RE 2015, Q3.a** [5 Marks]
- *"Define the term 'Artificial Intelligence'. Explain how AI techniques improve real-world problem solving."* — **GTU Winter-19, Q.8** [3 Marks]
- *"What is AI? State various characteristics of AI."* — **GTU Summer-20, Q.7** [4 Marks]

Marks-Scaled Model Answers (Group 1.1)

[CONCISE 2–3-MARK ANSWER]

Definition: John McCarthy (1956) defined Artificial Intelligence as *'the science and engineering of making intelligent machines, especially intelligent computer programs.'* Russell & Norvig define AI as the study of **rational agents** that perceive their environment and take actions that maximize their expected success.
AI Technique: An AI technique is a method that organizes and uses knowledge efficiently such that it: (1) captures generalizations, (2) is easily modified, and (3) can be used in situations even if inaccurate or incomplete.
Indicative Marking Scheme: Definition (1.5 Marks) + AI Technique / Key Characteristics (1.5 Marks).

[STRUCTURED 4–5-MARK ANSWER]

1. Formal Definitions of AI:
• *John McCarthy (1956):* The science and engineering of making intelligent machines.
• *Elaine Rich & Kevin Knight (1991):* The study of how to make computers do things at which, at the moment, people are better.
• *Russell & Norvig (AIMA):* The designing of rational agents that act to achieve optimal expected outcomes.

2. Characteristics of AI Problems:
• **High State Space Complexity:** AI problems (e.g., Chess, Medical Diagnosis) have combinatorial state explosion that brute force cannot solve.
• **Non-Algorithmic & Heuristic-Based:** Cannot be solved by deterministic step-by-step mathematical formulas; require heuristic search.
• **Handling Incomplete & Noisy Data:** Operates under real-world uncertainty using probabilistic reasoning.
• **Knowledge-Intensive:** Requires large structured domain knowledge bases rather than simple raw numeric computations.

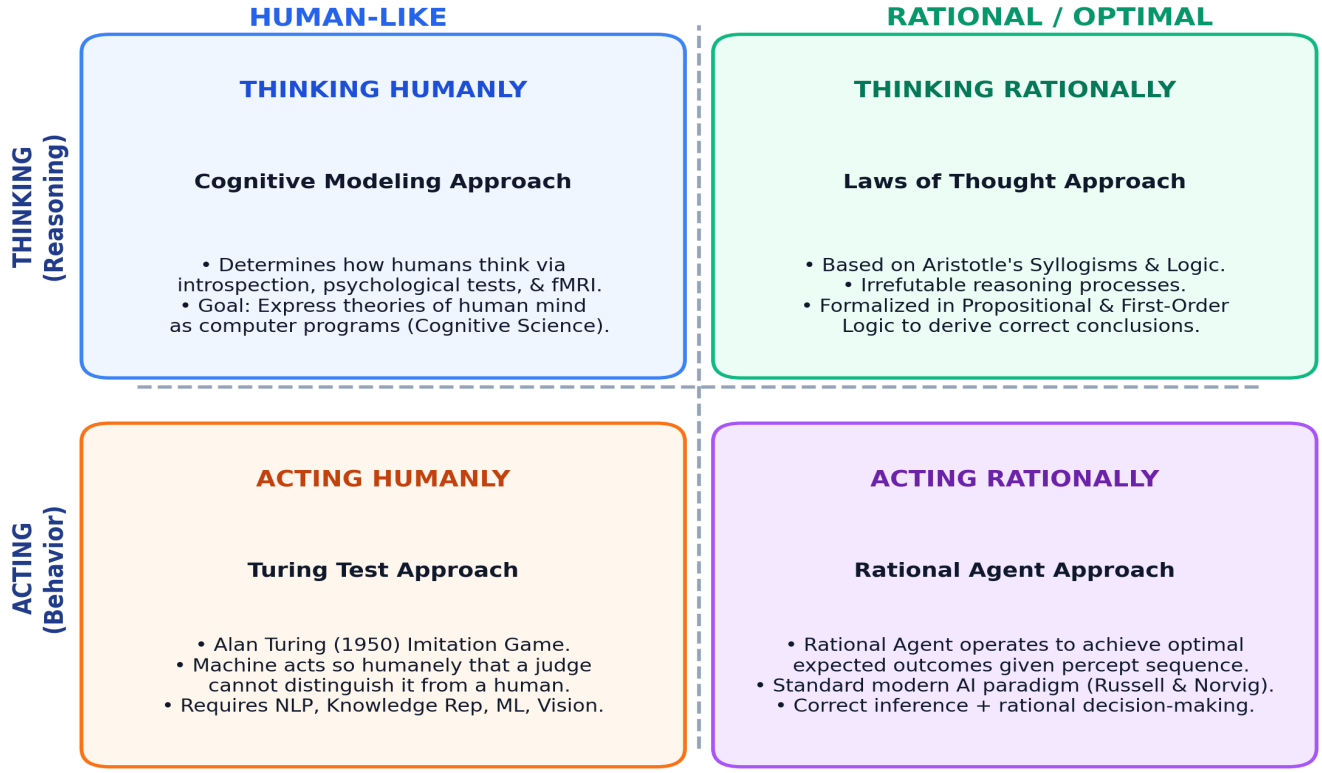
Indicative Marking Scheme: Standard Definitions (2 Marks) + 4 Characteristics explained with examples (3 Marks).

[DETAILED 7–10-MARK ANSWER]

1. Introduction & Formal Definitions:
Artificial Intelligence is an interdisciplinary field spanning Computer Science, Mathematics, Neuroscience, Logic, and Cognitive Psychology. It focuses on creating systems capable of human-level perception, reasoning, learning, and decision-making.

2. The Four Approaches to AI (2x2 Matrix):
As formalized by Russell & Norvig, AI definitions fall into four distinct quadrants based on human performance vs. rationality and thought vs. behavior:

Figure 1.1: The Four Approaches to Artificial Intelligence



3. Seven Core Characteristics of AI Problems:

1. *Is the problem decomposable?* Can it be broken into independent sub-problems (e.g., symbolic integration)?
2. *Can solution steps be undone?* Ignorable (theorem proving), Recoverable (8-puzzle), or Irrecoverable (chess/playing card)?
3. *Is the problem universe predictable?* Deterministic (chess) vs. Stochastic/Uncertain (medical diagnosis, taxi driving).
4. *Is a good solution absolute or relative?* Any valid solution vs. optimal path (e.g., Traveling Salesperson Problem).
5. *Is knowledge data-heavy or rule-heavy?* Requires massive factual knowledge bases vs. pure procedural search.
6. *Is human interaction required?* Conversational tutoring vs. autonomous batch processing.
7. *Is knowledge internal or external?* Static rule base vs. real-time dynamic environmental perception.

Indicative Marking Scheme: Definitions & Matrix (3 Marks) + Figure 1.1 Diagram (2 Marks) + 7 Characteristics Detailed (4 Marks) + Conclusion (1 Mark).

PYQ GROUP 1.2: The Turing Test and Total Turing Test

Repetition Frequency: *Asked 4 times (MU May 2016 | GTU 2017, 2018, 2019)*

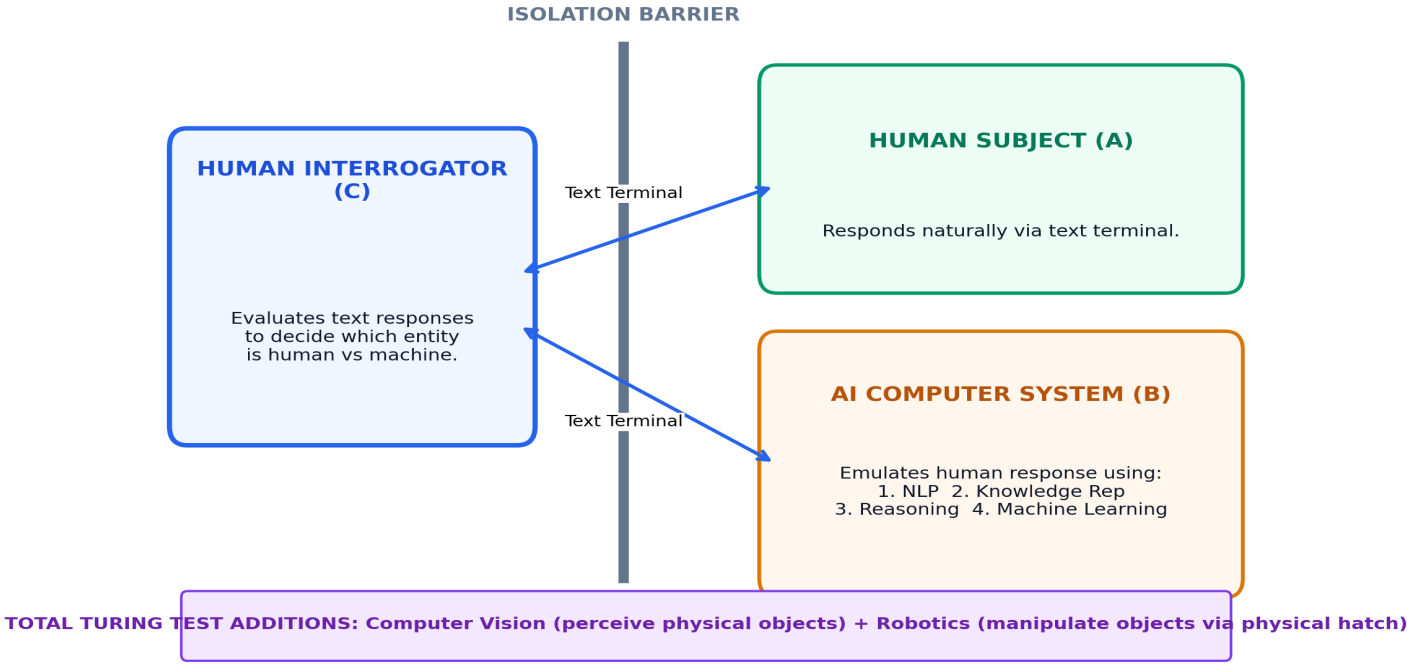
- *"Explain Turing test designed for satisfactory operational definition of AI."* — **MU May-16** [5 Marks]
- *"Discuss Turing test."* — **GTU Winter-17, Q.4 / GTU Summer-18, Q.5 / GTU Winter-18, Q.3** [3 to 4 Marks]
- *"What is the significance of the 'Turing Test' in AI? Explain how it is performed."* — **GTU Winter-19, Q.9** [4 Marks]

Marks-Scaled Model Answers (Group 1.2)

1. Concept & Significance: Proposed by Alan Turing (1950) in *'Computing Machinery and Intelligence'*, the Turing Test provides a satisfactory operational definition of intelligence based on **acting humanly** (the Imitation Game).

2. Test Setup & Architecture:
A human interrogator (C) is isolated from a human subject (A) and a computer system (B). Communication occurs via text terminals. If C cannot reliably distinguish the computer from the human after a 5-minute interrogation, the machine passes.

Figure 1.2: Turing Test and Total Turing Test Architecture



3. Four Capabilities Required for Standard Turing Test:
- *Natural Language Processing (NLP):* To communicate successfully in human language.
- *Knowledge Representation:* To store what it knows or hears.
- *Automated Reasoning:* To use stored knowledge to answer questions and draw new conclusions.
- *Machine Learning:* To adapt to new circumstances and detect patterns.

4. Total Turing Test Additions: Includes a physical hatch to test perceptual and manipulation abilities via:
- *Computer Vision:* To perceive physical objects.
- *Robotics:* To manipulate objects and move in physical space.

Indicative Marking Scheme: Concept (1 Mark) + Diagram Setup (1.5 Marks) + 4 Capabilities (1.5 Marks) + Total Turing Test Additions (1 Mark).

2.0 TOPIC 2: MODELING, INFERENCE, LEARNING & SUBSETS OF AI (AI vs ML vs DL)

PYQ GROUP 2.1: Machine Learning vs. Deep Learning & AI Relationship

Repetition Frequency: *Asked 2 times (NMIMS Jan 2023, Nov 2024)*

- *"Machine learning and Deep learning"* — **SVKM's NMIMS Re-Exam Jan 2023, Q7.A** [4 Marks]
- *"Discuss suitable examples for each of the following learning types - i. Supervised Learning, ii. Unsupervised Learning."* — **SVKM's NMIMS Final Exam Nov 2024, Q1.d** [5 Marks]

Marks-Scaled Model Answers (Group 2.1)

1. Three Core Pillars of Intelligent Systems:

• **Modeling:** Formalizing real-world knowledge into mathematical or symbolic structures (e.g., Propositional Logic, Semantic Networks).

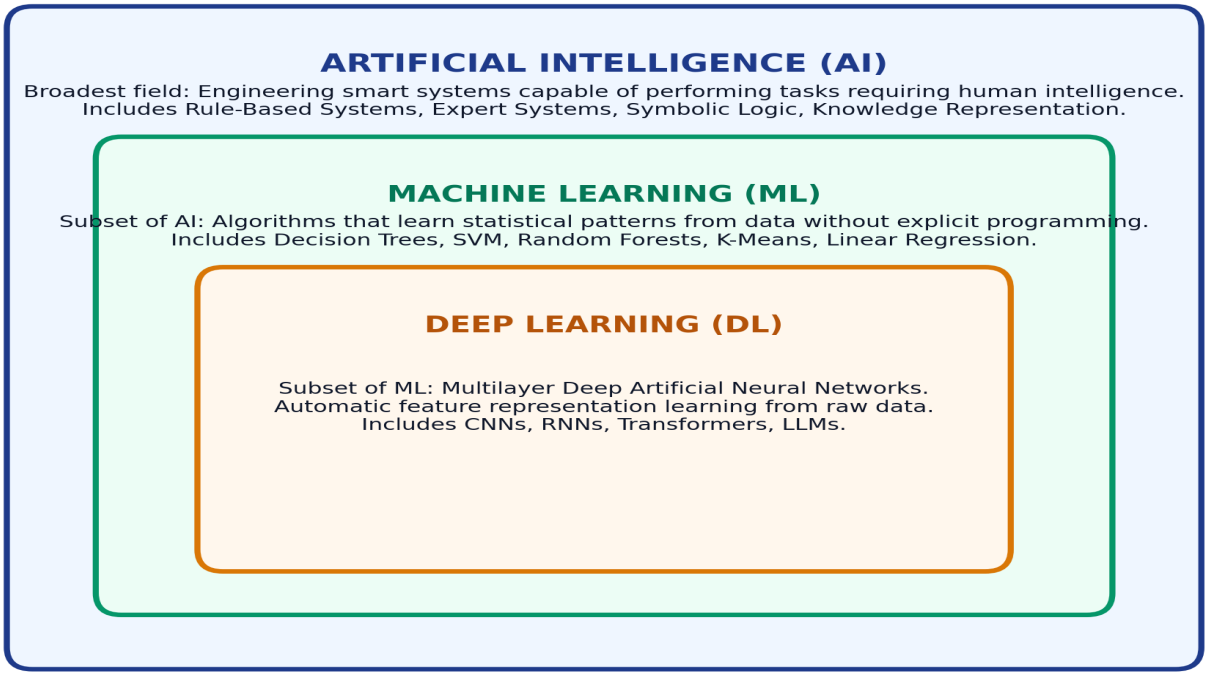
• **Inference:** Deriving new facts or conclusions from existing models using reasoning engines (e.g., Forward/Backward Chaining, Resolution).

• **Learning:** Adapting model parameters automatically based on environmental experience and training data.

2. Nested Hierarchy Relationship:

Artificial Intelligence is the overarching discipline. Machine Learning is a subset of AI, and Deep Learning is a specialized subset of ML.

Figure 2.1: Relationship Among AI, Machine Learning, and Deep Learning



Comparative Analysis: AI vs. Machine Learning vs. Deep Learning

Dimension	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Scope & Definition	Overarching field to build smart human-like machines.	Subset of AI learning patterns from data.	Subset of ML using multi-layer neural networks.
Feature Engineering	Hand-crafted symbolic rules & logic.	Manual feature extraction by domain experts.	Automated end-to-end feature representation learning.
Data Dependency	Operates on small rule bases & knowledge graphs.	Requires medium structured tabular datasets.	Requires massive unstructured data (images, text, audio).
Hardware Requirements	Standard CPU execution.	Standard CPU / GPU processing.	Heavy GPU / TPU parallel compute clusters.
Supervised / Unsupervised	Rule-based expert systems (MYCIN).	Decision Trees, SVM, K-Means Clustering.	CNNs (ResNet), Transformers (GPT-4), Autoencoders.

3.0 TOPIC 3: APPLICATIONS OF ARTIFICIAL INTELLIGENCE

PYQ GROUP 3.1: Task Domains & Real-World Applications of AI

Repetition Frequency: *Asked 4 times (MU 2023 | GTU 2012, 2018 | MDU 2018)*

- *"Explain Applications Of AI In Field Of Healthcare?"* — **MU M23** [5 Marks]
- *"Define and discuss different task domains / application areas of artificial intelligence."* — **GTU Winter-18, Q.6 / MDU BCA 2018, Q2.b** [3 to 5 Marks]
- *"Enlist and discuss the major task domains of Artificial Intelligence."* — **GTU Summer-12, Q.10** [7 Marks]

Marks-Scaled Model Answers (Group 3.1)

1. Healthcare & Medical Systems: Medical imaging analysis (X-ray, MRI tumor detection via CNNs), automated diagnostic rule-based expert systems (e.g., MYCIN), surgical robotic assist systems, and genomic drug discovery models.
2. Autonomous Transportation: Self-driving vehicles (Tesla, Waymo) utilizing computer vision, LiDAR, Radar, and sensor fusion for real-time path planning.
3. Finance & Banking: Fraud detection algorithms, automated credit scoring, algorithmic high-frequency trading, and conversational banking chatbots.
4. E-Commerce & Recommendation Engines: Personalized product recommendations (Amazon, Netflix) using collaborative filtering and deep retrieval models.
5. Natural Language Processing & Virtual Assistants: Speech recognition, automated translation (Google Translate), and large language model assistants (ChatGPT, Siri).
6. Industrial Automation & Robotics: Warehouse part-picking robots, automated quality inspection systems, and predictive equipment maintenance.

Indicative Marking Scheme: 5 major domains detailed with real-world examples (1 Mark per domain = 5 Marks total).

4.0 TOPIC 4: INTELLIGENT AGENTS, RATIONALITY & STRUCTURE

PYQ GROUP 4.1: Intelligent Agent, Rationality & Structural Equations

Repetition Frequency: *Asked 4 times (NMIMS Nov 2023 | MU Dec 2024 | GTU 2018)*

- *"Describe An Intelligent Agent With Diagram?"* — **MU D24** [5 Marks]
- *"What is an Intelligence Agent? Explain the concept of Learning Based Agent with the help of diagram."* — **SVKM's NMIMS Final Exam Nov 2023, Q5.a** [10 Marks]
- *"What is an ideal rational agent? Explain the concept of rationality."* — **GTU Summer-18, Q.13** [4 Marks]
- *"Define in your own words: Agent, Agent function, Agent program, Autonomy."* — **GTU / R&N; Chapter 2 Exercises** [4 Marks]

Marks-Scaled Model Answers (Group 4.1)

1. Definitions & Formal Equations:

• **Agent:** Anything that perceives its environment through **sensors** and acts upon that environment through **actuators**.

• **Percept Sequence (P^*):** The complete history of everything the agent has perceived from initialization to the present moment.

• **Agent Function (f):** An abstract mathematical mapping from percept sequence histories to actions:

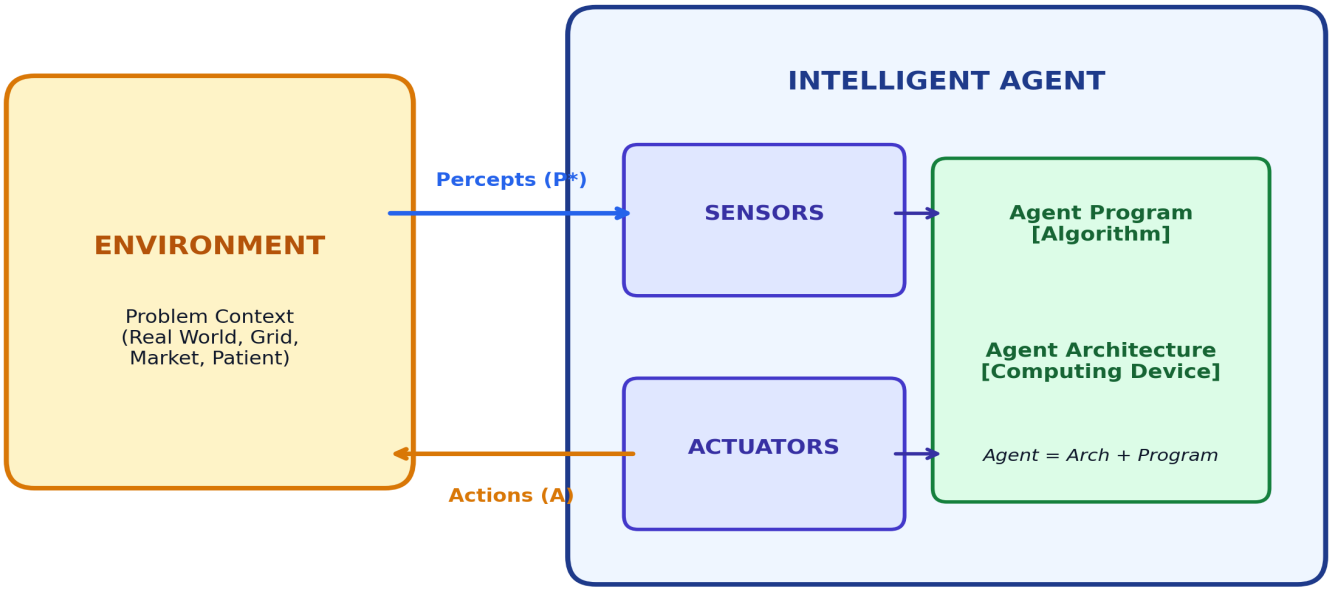
$$f: P^* o A$$• **Agent Program:** The concrete algorithm running on the physical architecture that implements f.

• **Fundamental Structural Relationship:**

$$\mathbf{Agent = Architecture + Agent\ Program}$$ where *Architecture* is the computing device with hardware sensors/actuators, and *Agent Program* is the software algorithm.

2. Agent-Environment Interaction Diagram:

Figure 3.1: Complete Agent-Environment Interaction Cycle



3. Concept of Rationality vs. Omniscience:

A **rational agent** is one that does the 'right thing'—specifically, for each possible percept sequence, a rational agent selects an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Four Governing Criteria of Rationality:

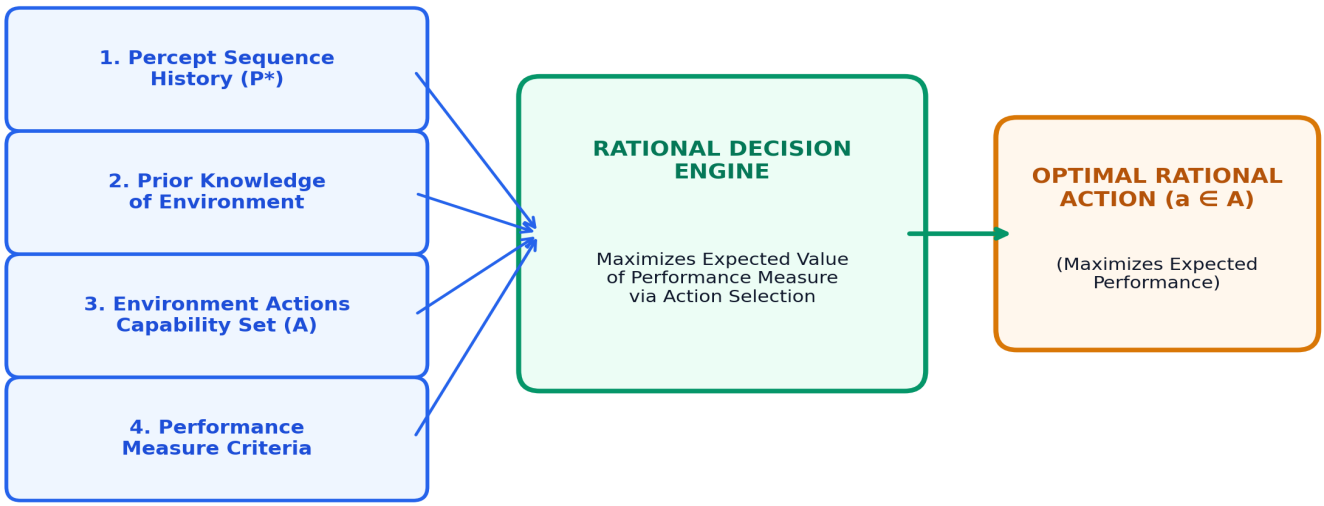
1. The *Performance Measure* defining criterion of success.
2. The agent's *Prior Knowledge* of the environment.

3. The *Actions* that the agent can perform.
4. The agent's *Percept Sequence (P^*)* to date.

Rationality vs. Omniscience & Perfection:
• *Omniscience:* An omniscient agent knows the **actual outcome** of its actions in advance (impossible in real stochastic worlds).
• *Rationality:* Rationality maximizes **expected performance** based on past percepts.
• *Autonomy:* A rational agent must learn what it can to compensate for partial or incorrect prior knowledge.

4. Rational Agent Decision Flow:

Figure 3.2: Rational Agent Decision Optimization Flow



5.0 TOPIC 5: PEAS FRAMEWORK & REAL-WORLD AGENT SPECIFICATIONS

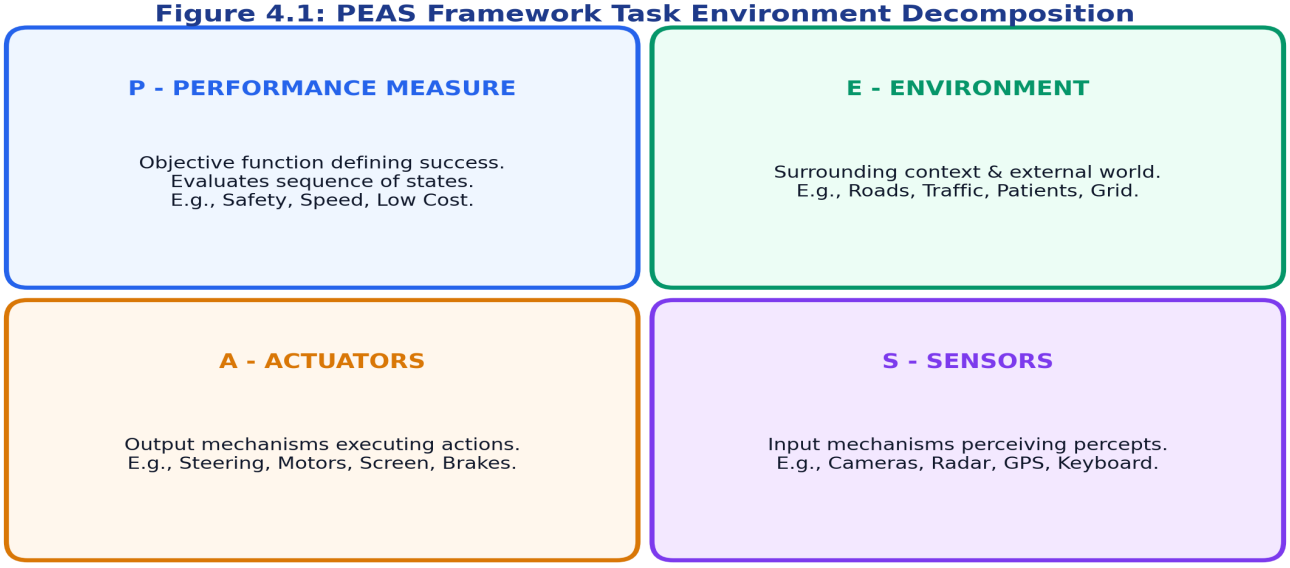
PYQ GROUP 5.1: PEAS Specifications for Real-World Intelligent Agents

Repetition Frequency: *Asked 8 times (NMIMS 2022–2026 | MU 2013–2025)*

- *"Explain the PEAS properties in context of Medical Diagnostic system."* — **SVKM's NMIMS Final Exam Nov 2023, Q1.b** [5 Marks]
- *"Write down the PEAS for the following intelligent agents - i. Self Driving Car, ii. A robotic Doctor who performs surgery on the patients"* — **SVKM's NMIMS Final Exam Nov 2024, Q2.b** [5 Marks]
- *"Define the PEAS properties and construct a complete PEAS description for an 'AI powered drone for environment monitoring', clearly explaining each component."* — **SVKM's NMIMS Final Exam Nov 2025, Q2.b** [5 Marks]
- *"Define the PEAS properties and construct a complete PEAS description for an 'AI-based Smart Refrigerator', clearly explaining each component."* — **SVKM's NMIMS Re-Exam Jan 2026, Q2.b** [5 Marks]
- *"Write PEAS For English Speaking Tutor?"* — **MU M23, M25** [5 Marks]
- *"What are PEAS descriptors? Give PEAS descriptors for Part-picking Robot."* — **MU May-13, Dec-14** [3 Marks]
- *"Give PEAS description for a robot soccer player. Characterize its environment."* — **MU May-16** [5 Marks]
- *"Discuss Wumpus World problem with its PEAS properties."* — **SVKM's NMIMS Final Exam Nov 2022, Q6.B** [10 Marks]

PEAS Framework Decomposition Diagram & Master Table

Figure 4.1: PEAS Framework Task Environment Decomposition



Agent Domain	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Self-Driving Car	Safety, speed, legal driving, passenger comfort, max profit.	Roads, traffic, pedestrians, weather, road signs.	Steering wheel, accelerator, brake, signal, horn.	Cameras, Radar, LiDAR, GPS, speedometer, sonar.
Medical Diagnostic	Healthy patient, correct diagnosis, min cost/lawsuits.	Patient, hospital staff, laboratory test reports.	Screen display (questions, test orders, diagnosis, therapy).	Keyboard / touch entry (symptoms, lab readings, history).
Surgical Doctor Robot	Precise incisions, organ safety, zero infection, rapid recovery.	Patient body, operating theater, surgical team.	Robotic arms, scalpels, cauterizers, suction tubes.	High-def 3D cameras, force-feedback haptic sensors.
AI Monitoring Drone	Area coverage, leak detection accuracy, battery efficiency.	Airspace, forest/wildlife, weather, terrain.	Propeller motors, camera gimbal, alert transmitter.	RGB camera, thermal infrared sensor, gas sensors.
Smart Refrigerator	Food freshness, zero waste, energy efficiency, auto-restock.	Refrigerator interior, food items, home network.	Cooling compressor, door display, Wi-Fi ordering API.	Internal optical cameras, RFID tags, weight sensors.
English Tutor Agent	Student score improvement, fluency, grammar, engagement.	Set of students, curriculum database.	Display screen (exercises, corrections), audio speaker.	Keyboard, mouse, microphone.
Part-Picking Robot	% Parts in correct bin, speed, zero component damage.	Conveyor belt, parts bin, warehouse workspace.	Jointed robotic arm, vacuum gripper, mechanical fingers.	2D/3D vision cameras, infrared proximity sensors.
Wumpus World Agent	Gold retrieved (+1000), avoiding pit/Wumpus (-1000), -1/step.	4x4 grid of rooms, Wumpus, pits, gold.	Turn Left, Turn Right, Move Forward, Grab, Shoot.	Stench, Breeze, Glitter, Bump, Scream.

6.0 TOPIC 6: TASK ENVIRONMENTS & 7 ENVIRONMENT DIMENSIONS

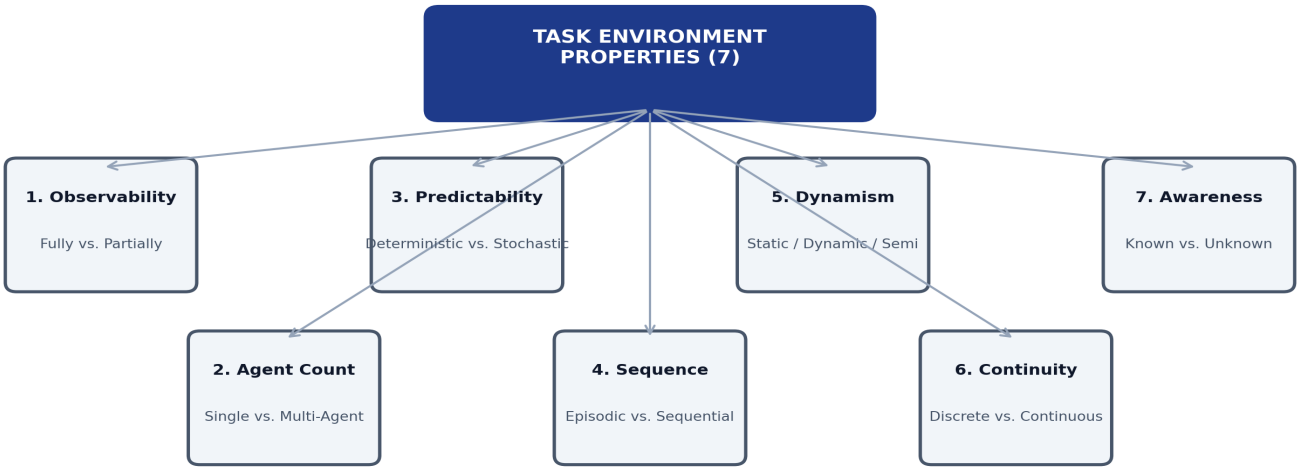
PYQ GROUP 6.1: Task Environment Properties & Verified Matrix

Repetition Frequency: *Asked 3 times (NMIMS Jan 2023 | GTU 2018 | MU 2013–2015)*

- *"Explain different types of environment with suitable example?"* — **SVKM's NMIMS Re-Exam Jan 2023, Q2.A** [10 Marks]
- *"Explain properties of environment."* — **GTU Summer-18, Q.14 / MU May-13, Dec-14** [4 to 5 Marks]
- *"Describe different types of environments applicable to AI agents."* — **MU Dec-13, May-15** [10 Marks]

Classification Tree & Verified 8-Environment Matrix

Figure 4.2: Task Environment Dimensions & Classification Tree



Task Environment	Observability	Agents	Determinism	Structure	Dynamism	Continuity
Chess (with clock)	Fully Observable	Multi-Agent	Deterministic	Sequential	Semi-Dynamic	Discrete
Crossword Puzzle	Fully Observable	Single-Agent	Deterministic	Sequential	Static	Discrete
Automated Taxi	Partially Observable	Multi-Agent	Stochastic	Sequential	Dynamic	Continuous
Medical Diagnosis	Partially Observable	Single-Agent	Stochastic	Sequential	Dynamic	Continuous
Part-Picking Robot	Partially Observable	Single-Agent	Stochastic	Episodic	Dynamic	Continuous
Interactive Tutor	Partially Observable	Single-Agent	Stochastic	Sequential	Dynamic	Discrete
Internet Shopping	Partially Observable	Single-Agent	Stochastic	Sequential	Dynamic	Discrete

Task Environment	Observability	Agents	Determinism	Structure	Dynamism	Continuity
Vacuum Cleaner World	Fully / Partially	Single-Agent	Deterministic	Sequential	Static	Discrete

7.0 TOPIC 7: AGENT ARCHITECTURES & 5 AGENT TYPES

PYQ GROUP 7.1: Overview & Individual Agent Architectures

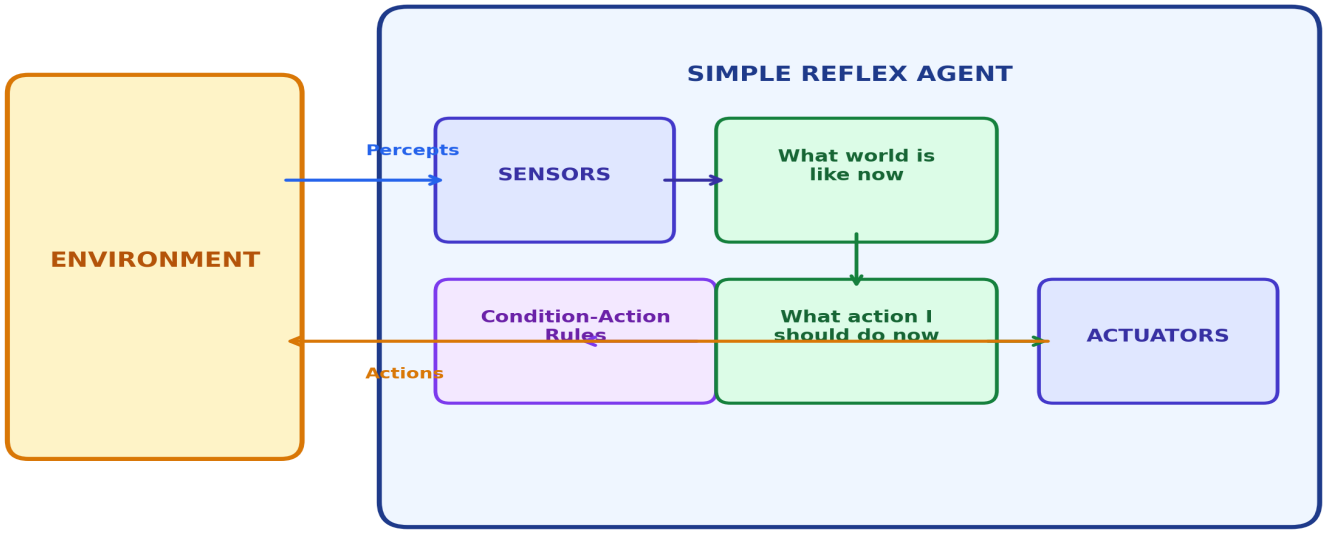
Repetition Frequency: *Asked 8 times (NMIMS 2022–2026 | MU 2014–2015)*

- *"Explain different types of agents with suitable examples?"* — **SVKM's NMIMS Final Exam Nov 2022, Q2.A** [10 Marks]
- *"Explain various types of intelligent agents, state limitations of each and how it is overcome in other type of agent."* — **MU May-14, Dec-15** [10 Marks]
- *"Explain the architecture of a Learning-based Agent with neat diagram. Clearly describe the roles of its components and showing how the interaction between these components improves decision-making over time."* — **SVKM's NMIMS Final Exam Nov 2025, Q2.a / Nov 2024, Q2.a / MU Dec-15** [5 to 10 Marks]
- *"Explain the architecture of a Goal-based Agent with neat diagram. Clearly describe the roles of its components."* — **SVKM's NMIMS Re-Exam Jan 2026, Q2.a / Feb 2025, Q2.a / MU May-15** [5 to 10 Marks]

1. Simple Reflex Agent Architecture

Definition: Selects actions based *only* on the current percept, ignoring the rest of the percept history.
Condition-Action Rule: $if\ \text{car-in-front-is-braking}\ then\ \text{initiate-braking}$.
Correction on Observability: Deterministic simple reflex agents fail in partially observable worlds due to infinite loops. However, *randomized / stochastic action selection* allows simple reflex agents to break infinite loops.

Figure 5.1: Simple Reflex Agent Architecture

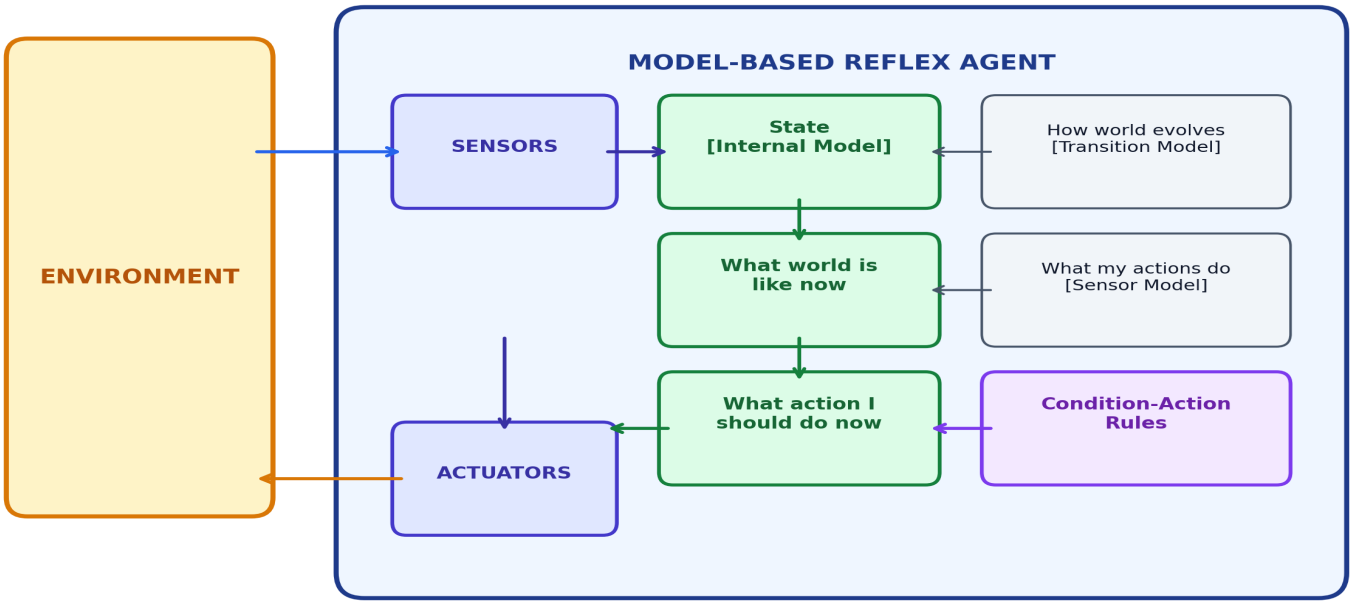


2. Model-Based Reflex Agent Architecture

Definition: Maintains an internal **State** to keep track of unobserved aspects of the current environment.

- **Transition Model:** Knowledge about *'how the world evolves'* independently of the agent.
- **Sensor Model:** Knowledge about *'how the world visible state maps to sensors'*.

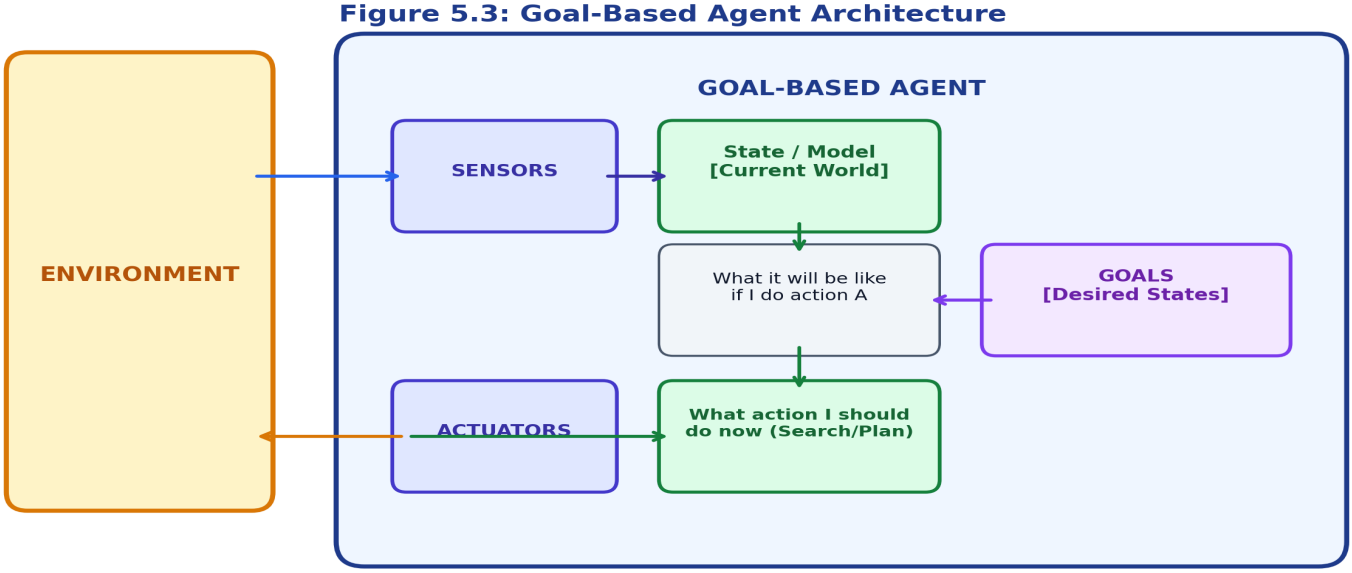
Figure 5.2: Model-Based Reflex Agent Architecture



3. Goal-Based Agent Architecture

Definition: Combines internal state information with explicit **Goal** descriptions that define desirable situations.
Uses search and planning algorithms to evaluate *'what it will be like if I do action A'*.

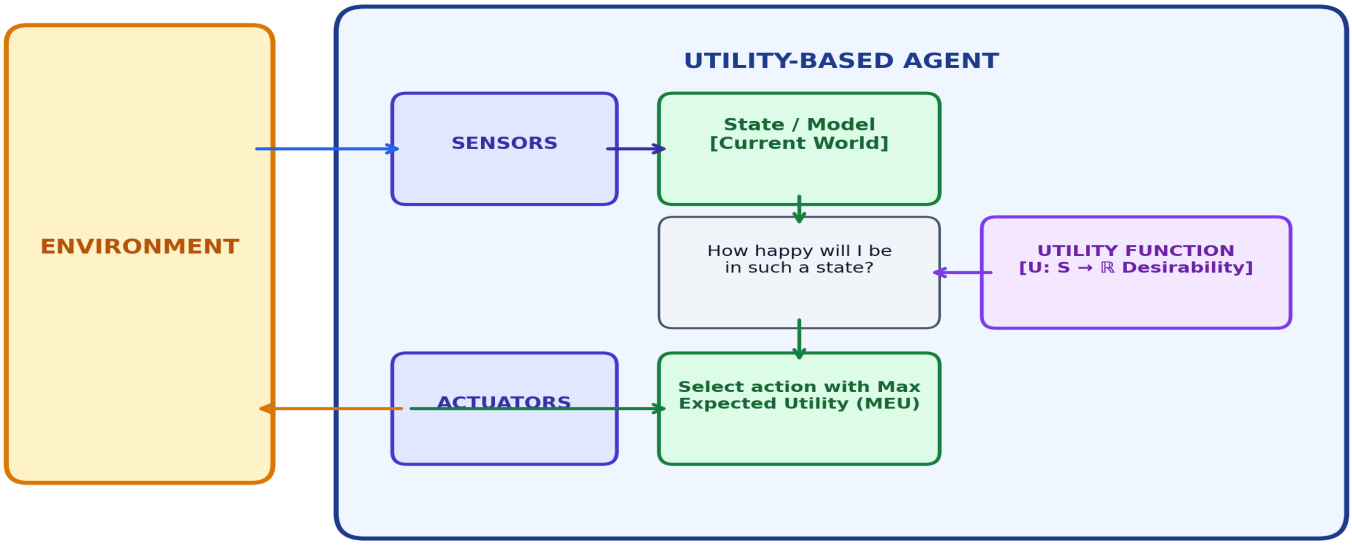
Figure 5.3: Goal-Based Agent Architecture



4. Utility-Based Agent Architecture

Definition: Uses an internal **Utility Function ($U: S \to \mathbb{R}$)** to measure degree of happiness or desirability of a state. Enables trade-offs when goals conflict or carry uncertainty via Max Expected Utility (MEU).

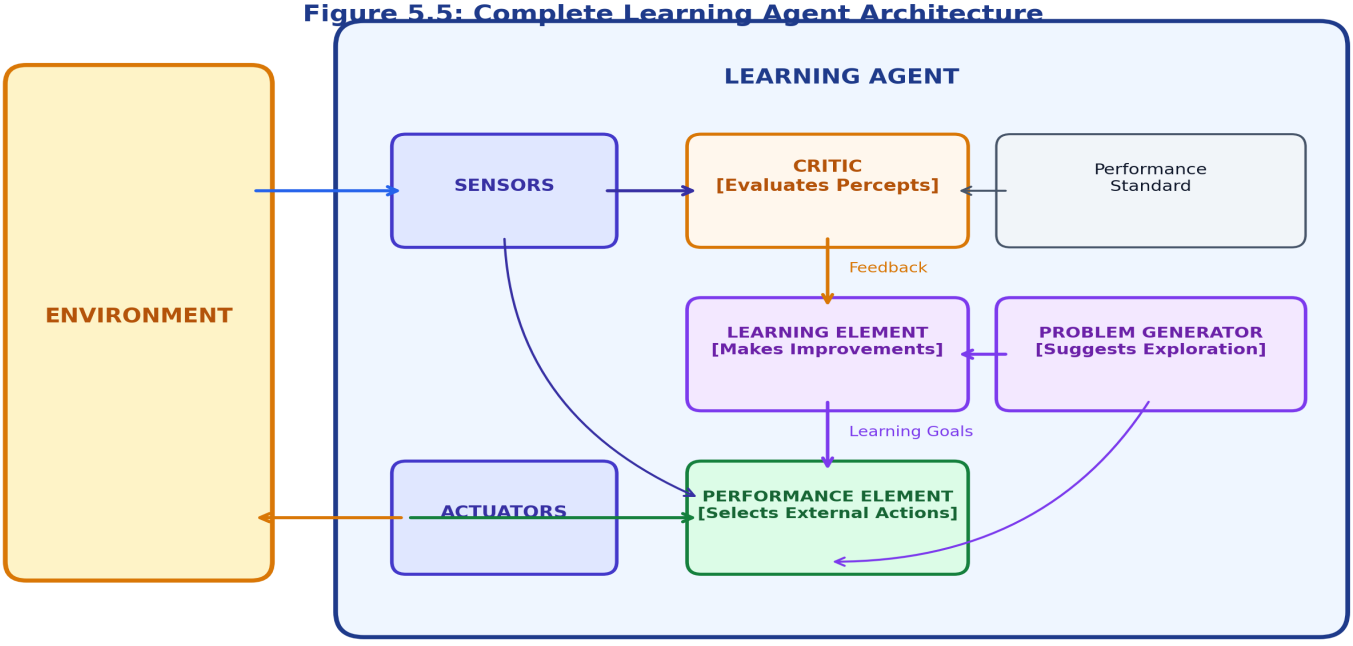
Figure 5.4: Utility-Based Agent Architecture



5. Learning Agent Architecture

Definition: Conceptual framework separated into four distinct operational components:
1. **Performance Element:** Selects external actions based on percepts.
2. **Critic:** Evaluates agent actions against a fixed *Performance Standard* and generates feedback.
3. **Learning Element:** Uses Critic feedback to make improvements to the Performance Element.
4. **Problem Generator:** Suggests exploratory actions (*Exploration*) to discover new experiences.

Figure 5.5: Complete Learning Agent Architecture

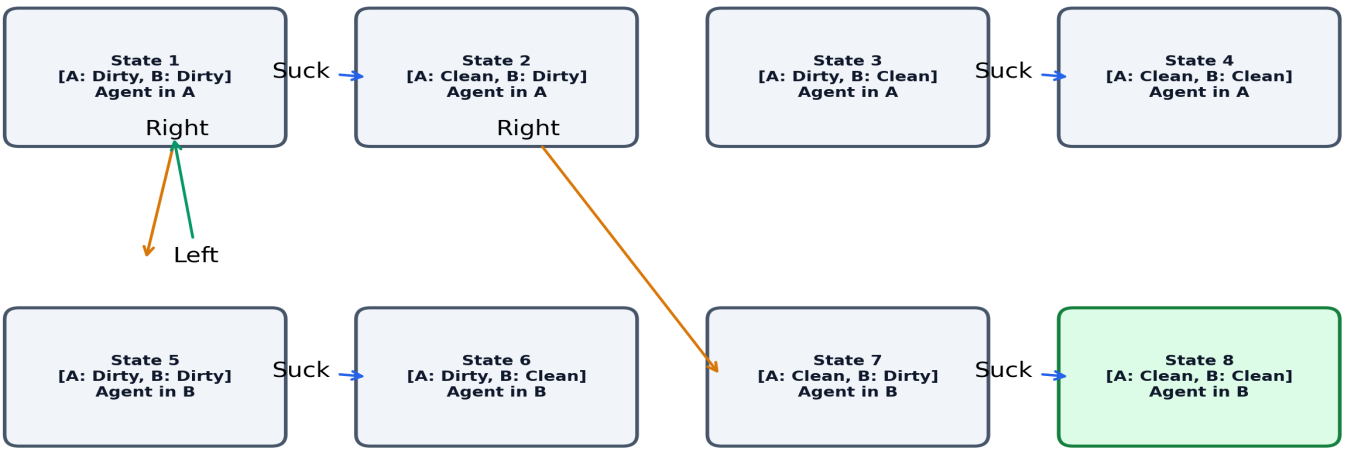


8.0 TOPIC 8: VACUUM-CLEANER WORLD CASE STUDY

1. State Space Formalization:
• Locations: 2 (Location A, Location B).
• Dirt Status per Location: 2 (Clean, Dirty).
• Total World States: $2 \text{ locations} \times 2^2 \text{ dirt states} = \mathbf{8\ \text{possible states}}$.
• Actions: $\{\text{Left}, \text{Right}, \text{Suck}, \text{NoOp}\}$.

2. Complete 8-State Space Transition Graph:

Figure 6.1: Complete Vacuum-Cleaner World 8-State Space Transition Diagram



2 Locations × 2^2 Dirt States = 8 Possible World States | Goal State: Clean Floor (State 4 or State 8)

9.0 SYLLABUS TOPICS WITHOUT AVAILABLE PYQs (PROBABLE QUESTIONS)

PROBABLE QUESTION 9.1: Hard Computing vs. Soft Computing [5 Marks]
Question: Differentiate between Hard Computing and Soft Computing in Artificial Intelligence.

Model Answer:
• **Hard Computing:** Based on crisp 0/1 deterministic logic, strict mathematical algorithms, and precise binary inputs. Fails under noisy or incomplete data.
• **Soft Computing:** Based on computational intelligence (Fuzzy Logic, Artificial Neural Networks, Genetic Algorithms). Tolerates imprecision, uncertainty, and partial truth.
• **Comparison Table:** Hard Computing uses Boolean Logic & crisp outputs; Soft Computing uses Fuzzy/Probabilistic Logic & approximate solutions.

10.0 INDEXES, FREQUENCY ANALYSIS & SYLLABUS CHECKLIST

FINAL SYLLABUS COMPLETENESS CHECKLIST:

- [X] Definitions of AI (John McCarthy, Rich & Knight, Russell & Norvig) verified.
- [X] Four Approaches to AI (Thinking/Acting Humanly/Rationally) & 2x2 Matrix verified.
- [X] Turing Test & Total Turing Test (6 required capabilities) verified.
- [X] Modeling, Inference, Learning & AI-ML-DL Hierarchy verified.
- [X] Intelligent Agents, Percepts (P^*), Agent Function $f: P^* \to A$, Agent Program, Architecture verified.
- [X] Concept of Rationality, Performance Measures, Autonomy, Omniscience distinction verified.
- [X] PEAS Framework & 8 Real-World Agent Master Tables verified.
- [X] 7 Task Environment Dimensions & Verified 8-Environment Matrix verified.
- [X] 5 Agent Architectures (Simple Reflex, Model-Based, Goal-Based, Utility-Based, Learning Agent) verified.
- [X] Vacuum-Cleaner World 8-State Space Transition Diagram & Analysis verified.